fn make_stable_group_by

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November 12, 2026

> This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `make_stable_group_by` in `mod.rs` at commit a105afe (outdated[1]).

1 Hoare Triple

Precondition

Compiler-verified

- Generic `M` implements `UnboundedMetric`.

Caller-verified

None

Pseudocode

```
def make_stable_group_by(
    input_domain: DslPlanDomain, input_metric: FrameDistance[M], plan: DslPlan
) -> Transformation[DslPlanDomain, DslPlanDomain, FrameDistance[M], FrameDistance[M]]:
    if M.ORDERED:
        raise "Sample-and-aggregate requires an unordered input metric"

    match plan:
        case DslPlan.GroupBy(input, keys, aggs, apply, maintain_order, options):
            pass
        case _:
            raise "Expected group-by in logical plan"

    if apply is not None:
        raise "apply is not currently supported"

    if options != GroupbyOptions.default():
        raise "options is not currently supported"

    t_prior = input.make_stable(input_domain, input_metric)
    middle_domain, middle_metric = t_prior.output_space()

    expr_domain = WildExprDomain(
        columns=middle_domain.series_domains,
        context=Context.RowByRow,
```

[1] See new changes with `git diff a105afe..9617bc23c rust/src/transformations/make_stable_lazyframe/group_by/mod.rs`

```
    )

    for key in keys:  #
        key.make_stable(expr_domain, L0PInfDistance(middle_metric[0]))

    for agg in aggs:  #
        check_infallible(agg, Resize.Allow)

    schema = Schema.from_columns(middle_domain.series_domains)
    for agg in aggs:  #
        check_permutation_invariance(agg, schema)

    if middle_metric[0].identifier().is_some():  #
        raise "identifier is not currently supported"

    # prepare output domain series
    output_schema = middle_domain.simulate_schema(  #
        lambda lf: lf.group_by(keys).agg(aggs)
    )
    series_domains = [SeriesDomain.new_from_field(f) for f in output_schema]

    # prepare output domain margins
    h_keys = set(keys)

    def output_margin(m: Margin) -> Margin:
        m = m.clone()
        m.by = materialize_grouping_keys(m.by)  #
        m.invariant = None
        return m

    margins = [
        output_margin(m) for m in middle_domain.margins if m.by.is_subset(h_keys)
    ]

    output_domain = FrameDomain.new_with_margins(series_domains, margins)

    def stability_map(d_in: Bounds) -> Bounds:
        #
        contributed_rows = d_in.get_bound(set()).per_group
        #
        contributed_groups = d_in.get_bound(h_keys).num_groups

        influenced_groups = option_min(contributed_rows, contributed_groups)
        if influenced_groups.is_none():
            return (
                "an upper bound on the number of contributed rows or groups is required"
            )

        per_group = influenced_groups.inf_mul(2)  #

        bound = Bound(by=set(), per_group=per_group, num_groups=None)
        return Bounds([bound])  #

    t_group_agg = Transformation.new(
        middle_domain,
        output_domain,
        lambda plan: DslPlan.GroupBy(
            input=plan,
            keys=keys,
            aggs=aggs,
            apply=None,
            maintain_order=maintain_order,
            options=options,
        ),
```

```
89          middle_metric,
90          middle_metric,
91          StabilityMap.new_fallible(stability_map),
92      )
93
94      return t_prior >> t_group_agg
95
96
97  # Called only for subsets of validated grouping expressions.
98  def materialize_grouping_keys(by: set[Expr]) -> set[Expr]:
99      return {col(key.meta().output_name()) for key in by}
```

Postcondition

Theorem 1.1. For every setting of the input parameters `(input_domain, input_metric, plan)` to `make_stable_group_by` such that the given preconditions hold,
`make_stable_group_by` raises an error (at compile time or run time) or returns a valid transformation. A valid transformation has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke`(x) and `invoke`(x') either both return the same error or neither return an error.

2. (Appropriate output domain). For every member x in `input_domain`, `function`(x) is in `output_domain` or raises a data-independent runtime error.

3. (Stability guarantee). For every pair of members x and x' in `input_domain` and for every pair $(\texttt{d_in}, \texttt{d_out})$, where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for
`output_metric`, if x, x' are `d_in`-close under `input_metric`, `stability_map(d_in)` does not raise an error, and `stability_map(d_in)` $=$ `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_metric`.

Appropriate Output Domain. By line 27 the grouping keys are stable row-by-row transformations of the data. By line 30 the aggregates are infallible. The check also excludes positional inputs and data-dependent quantile parameters, ensuring that aggregates depend only on their respective groups.

By the postcondition of `DslPlan.simulate_schema`, `series_domains` follows the expected schema of members in the output domain. The output column domains preserve the schema while allowing infallible black-box aggregation.

Only margins whose original grouping expressions are a subset of the grouping keys are preserved. On line 51, each preserved key becomes `col(output_name)`. Polars evaluates a grouping expression on the input and stores its value in the output column with that name. The translated margin therefore partitions output rows by the same labels as the original margin partitions input rows. Each input row contributes to at most one output row, so the upper bounds on group lengths and the number of groups remain valid. Re-evaluating the original expression after aggregation is not generally equivalent and must not inherit the original margin. Among these margins, invariants are discarded.

It has been shown that for every element x in `input_domain`, `function`(x) is in `output_domain` or raises a data-independent runtime exception. □

The stability guarantee doesn't attempt to claim the broadest set of possible bounds on output distances, rather it only claims a simple bound that might be useful for the user. This can be extended in the future, but is sufficient for select queries.

Stability guarantee. We first simplify the problem on line 37 by only considering datasets that differ by rows, not by identifiers.

Since ordered input metrics are rejected, row permutations have zero distance. The output metric is unchanged, so group ordering is immaterial. The only output distance bound derived is for the empty grouping set, namely the entire output table.

By line 34, accepted aggregations are invariant in distribution under input-row permutations. Their row-wise inputs commute with permutations, and order-sensitive reducers require a trusted uniform permutation, producing the same distribution regardless of input order. Grouping keys have canonical representatives, so unchanged groups can be coupled to have identical aggregate outputs.

We first retrieve optional upper bounds on the number of rows and groups an individual may contribute on lines 62 and 64. Both of these bounds directly correspond to the number of rows an individual may influence in the resulting dataset.

$$\max_{x \sim x'} d_{\text{FrameDistance<M>}}(\texttt{function}(x), \texttt{function}(x')) \tag{1}$$

$$\leq \text{option_min}(\texttt{contributed_rows}, \texttt{contributed_groups}) \cdot 2 \tag{2}$$

Since each influenced row accounts for one addition and one removal, the distance is twice the number of influenced rows. This is reflected on line 73. The resulting bound is returned on line 76, satisfying the postcondition. □